

Lab 6 Activity

PSYC 6802: Intro to Psychological Statistics

```
library(tidyverse)
```

1

In this activity, we will work with the `Prestige` data in the `carData` package.

```
data("Prestige", package = "carData")
```

Look at the help page to better understand these data. Here, we want to predict the `income` of occupational incumbents from the average `education` of occupational incumbents. Use `ggplot` to create a scatterplot with one or more best-fit curves. What type of relationship do you think best characterizes the relationship between these variables?

2

Fit a simple linear regression model predicting `income` from `education` and evaluate assumptions. What do you conclude?

3

Looking ahead: the `lm` function takes formula input like we used with `geom_smooth()`. Fit a regression equation for the best-fit quadratic/cubic curve predicting `income` from `education`. You can find the model residuals in the same way as we did for the linear model. Do the quadratic or cubic models do a better job meeting our assumptions about residuals? If not, can you think of what might be a better solution?